

VIDHI VATS

Software Development Engineer | Backend & Full-Stack | AI-Powered Systems

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PROFESSIONAL SUMMARY

Software Development Engineer and incoming M.Tech CS student at IIT Patna, skilled in building scalable, real-time distributed systems with React, Django, and Node.js on GCP/AWS. Experienced with high-throughput data pipelines using Redis and RabbitMQ, and AI-powered applications built on LangChain, RAG, and vector search. Strong foundation in data structures, system design, and clean backend architecture.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript

Backend: Django, Node.js, Express.js, Nest.js, Flask, REST APIs

Frontend: React.js, HTML, CSS, Tailwind, MUI

AI / LLM: LangChain, OpenAI APIs, RAG, Embeddings, Vector Stores, AI Agents, Prompt Engineering

Databases & Messaging: MongoDB, PostgreSQL, MySQL, Redis, RabbitMQ

Cloud & DevOps: GCP, AWS, Docker, Firebase, GitHub Actions, CI/CD

Core Concepts: Data Structures & Algorithms, OOP, Microservices, System Design, Distributed Systems

EXPERIENCE

Software Developer Intern — Data Sailors Private Limited

06/2025 – 02/2026

- Built an enterprise-grade real-time monitoring dashboard in React.js, improving data visibility across a 12-node distributed sensor network.
- Designed scalable backend services with Django + MongoDB to ingest and process high-frequency real-time data streams.
- Introduced Redis caching and a RabbitMQ message queue with WebSocket-driven live updates, decoupling ingestion from processing to smooth traffic spikes and keep the dashboard responsive under load.
- Engineered a Data Concentrator Unit on Raspberry Pi for secure data collection and transmission from edge devices.
- Deployed and managed infrastructure on Google Cloud Platform (GCP) with Docker and secure VPN + SSH access, ensuring scalability, reliability, and safe remote operation.
- Scaled the architecture from a single node to a distributed cluster, improving performance and fault tolerance.

PROJECTS

Personal Portfolio Website — React, Vite, JavaScript, EmailJS, Vercel

vidhivats-portfolio.vercel.app

- Designed and deployed a responsive personal portfolio featuring an EmailJS-powered contact form and an interactive resume chatbot.
- Automated build and deployment through a GitHub → Vercel CI/CD pipeline with environment-based secret management.

GenAI RAG Document Assistant — Python, LangChain, OpenAI, Vector Stores, Docker, CI/CD

- Built a Retrieval-Augmented Generation chatbot that answers natural-language questions over a custom document corpus using LangChain and OpenAI embeddings.
- Designed an ingestion pipeline that chunks documents, generates embeddings, and indexes them in a vector store for fast semantic retrieval and querying.
- Implemented context-aware query handling and prompt workflows to produce grounded, low-hallucination answers with source citations.
- Containerized the app with Docker and automated build/test/deploy with GitHub Actions (CI/CD), deploying on Firebase for high availability.

Network Vulnerability Scanner — React, Django, Python

- Built a full-stack security tool in React + Django to scan and analyze network vulnerabilities with real-time device discovery and scanning.
- Simulated real-world attacks (MITM, ARP spoofing, packet sniffing) and built Python brute-force modules for SSH/Telnet security testing.
- Optimized packet processing for real-time vulnerability detection.

CERTIFICATIONS

Full Stack Generative & Agentic AI with Python — Udemy (Hitesh Choudhary & Piyush Garg): LLMs, AI agents, RAG, and automation pipelines with hands-on projects.

EDUCATION

M.Tech, CSE — Indian Institute of Technology (IIT), Patna

2026 – 2028 (Pursuing)

B.Tech, Electrical Engineering — Shri Mata Vaishno Devi University

Jul 2021 – May 2025